XIAMEN UNIVERSITY MALAYSIA

ASSIGNMENT 2



Dr. Mohammadmahdi Ariannejad

mohammadmahdi.ariannejad@xmu.edu.my

Try to solve the questions **Completely with all steps**! (Effective for grading)

The answer can be in **ONE PDF** format file only (individual activity)

Cheating in any condition will **consider zero Mark** in any exam and assignment!

only **1 Attempt** is allowed.

The deadline is the **29th of June 2023 at 11:00 AM**

Late submission will impose **a penalty (communicate with me if delay happened)**!

Writing from solution manuals, web blogs, shared blogs, or any type of getting an answer will cause you zero marks!

Cheating and writing from **Internet resources** will consider as cheating, please avoid any kind of cheating! (**Do not share answers** at any cost!)

1. Writing a positive integer in binary means expressing it as a sum of powers of 2:

$$(b_{n-1}\ b_{n-2}\ \ldots\ b_2\ b_1\ b_0)_2 \equiv \sum_{i=0}^{n-1} b_i\ 2^i$$

This representation makes sense for unsigned numbers. But what about signed numbers? How can we represent a signed number as a sum of powers of 2? Which powers of two are represented and what are the b_i ?

2. Trade-offs in the choice of operands, seems that listed many design possibilities but have not focused on which approach has been adopted. In fact, there is no best choice, and each operand style in practice.

a) Why hasn't one particular style emerged as optimal?

b) Discuss several potential design goals, and explain how each relates to the choice of operands.

3. Computer designers have invented many variations to the basic form of microcode. For example, the CPU hardware implements the fetch-execute cycle and invokes a microcode procedure for each instruction. What is the advantage and disadvantages of microcode?

4. For the following assume that values A, B, C, D, E, and F reside in memory. Also assume that instruction operation codes (opcode) are represented in 8 bits, memory addresses are 64 bits, and register addresses are 6 bits.

Stack	Accumulator	Register (register-memory)	Register (load-store)
Push A	Load A	Load R1,A	Load R1,A
Push B	Add B	Add R3,R1,B	Load R2,B
Add	Store C	Store R3,C	Add R3,R1,R2
Pop C			Store R3,C

For each instruction set architecture shown in Figure above, how many addresses, or names, appear in each instruction for the code to compute C = A + B, and what is the total code size?

5. One of the interesting ideas about the assembly language is the characteristics of A Low-level Programming Language. The alternative to a high-level language is known as a low-level language, please mention the characteristics with explanations for each of characteristics (5 item).

6. As we know the Multiple Memories with Separate Controllers can be consider by architectures that may contain multiple physical memories.

a) Why does it help to have multiple memories, each with their own controller?

b) How the memory banks can use used with Von Neumann Architecture.

c) How do memory banks appear to a programmer?

7. Chen, Xie, and Jin are building custom hardware and compilers for your project, because they apparently have a lot of free time and like writing hardware and compilers. The following information is known.

• Xie has already gone ahead and built his design, and it has a known execution length for your new program of 600 seconds.
• Chen proposes a singe cycle data path, and plans to use a compiler that will generate 300 Billion dynamic instructions per execution.
• Jin proposes a pipelined data path, with a clock period of 1.2 ns. Her compiler will generate 400 Billion instructions. Of these 400 Billion instructions, 25% are loads, and 20% are branches. 40% of all loads cause a 2-cycle load-use stall. The penalty for a mis-predicted branch is 5 cycles. The processor has full bypassing, and does not suffer from any other stalls.

(i) How fast does Chen need to make his clock cycle in order to make the program run faster than Xie's?

(ii) How accurate does Jin's branch predictor need to be in order to make the program run faster than Xie's?

8. In the virtual memory;

a) By consideration for the multiprogramming, how we an archive to the Creating Virtual Spaces Dynamically (Examine technologies that have been used to create dynamic virtual memory systems. consider three examples, explain in details)?

b) What is the relationship Between Virtual Memory and Caching?

d) How can a cache be engineered to avoid the problem of ambiguity that occurs when multiple processes use the same range of addresses? (Please explain in details)

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